Investigating Double Descent

Multiple Regression Analysis on the Friedman Datasets

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AGENDA

1 Introduction

2 Background

3 Summary of the Data

4 Methods and Implementation

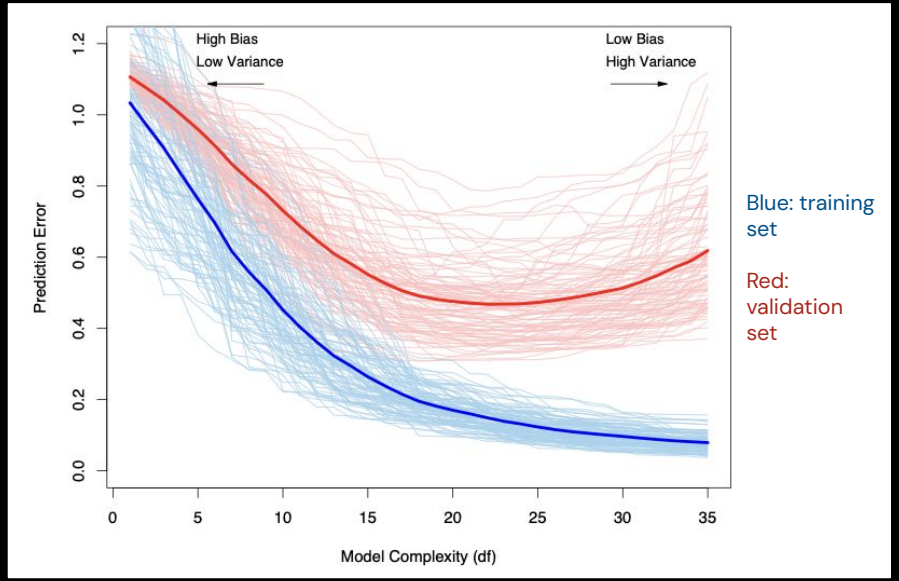
5 Results

6 Discussion

PROBLEM STATEMENT

Does double descent occur across multiple model classes on synthetic regression datasets, and under what conditions?

THE CLASSICAL VIEW



Blue: training set

Red: validation set

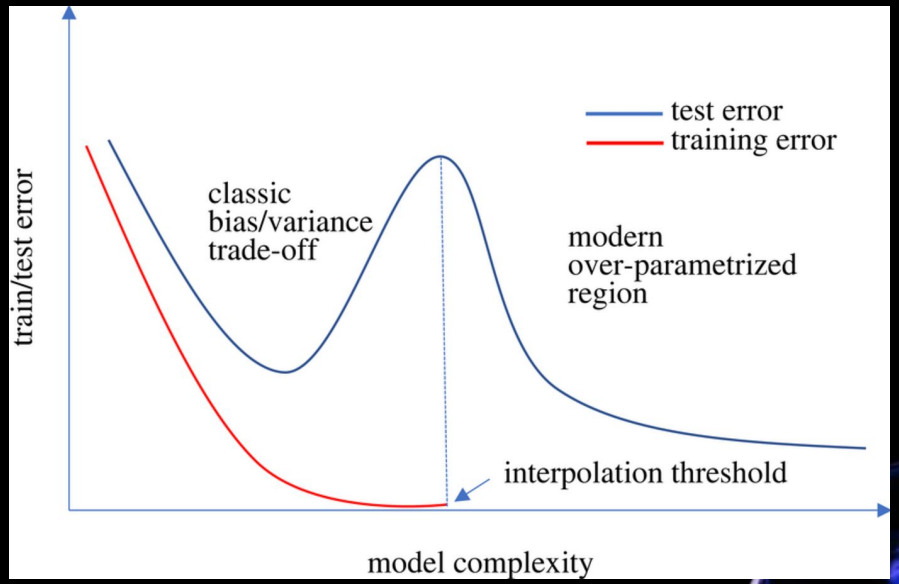
Small models underfit.

Large models overfit.

Prof. Olga Vitek

Classical statistical learning theory predicts a bias-variance tradeoff as model complexity increases.

DOUBLE DESCENT



Past the interpolation threshold, increases in model complexity can cause the test error to decrease again.

Medium

MODELS STUDIED

We investigate double descent across four model classes of increasing complexity:

01

Polynomial Regression

Varies degree

02

Random Feature Regression

Varies number of random features

03

Kernel Ridge Regression

Varies regularization strength

04

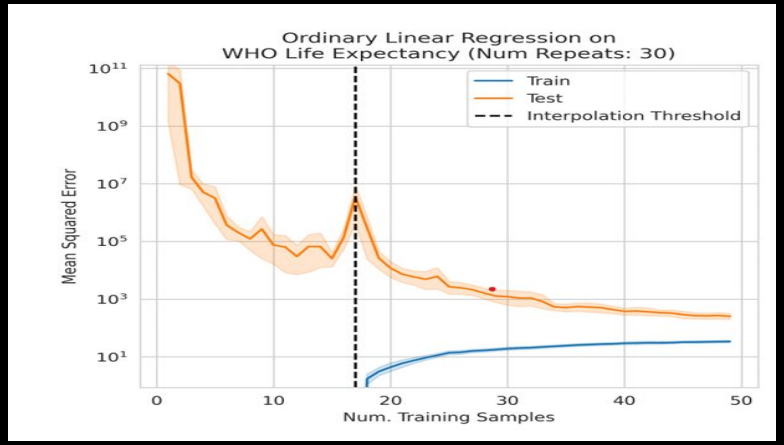
Neural Network

Varies number of hidden units

Each model is evaluated on three synthetic regression datasets.

LITERATURE OVERVIEW

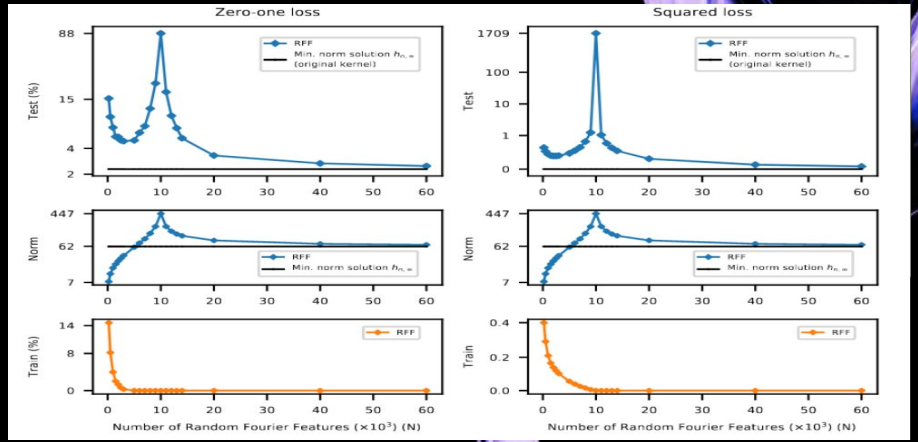
Polynomial Regression



Pros: Clear intuition with clean math that explains why double descent happens.

Cons: Limited to linear regression.

Random Features Regression



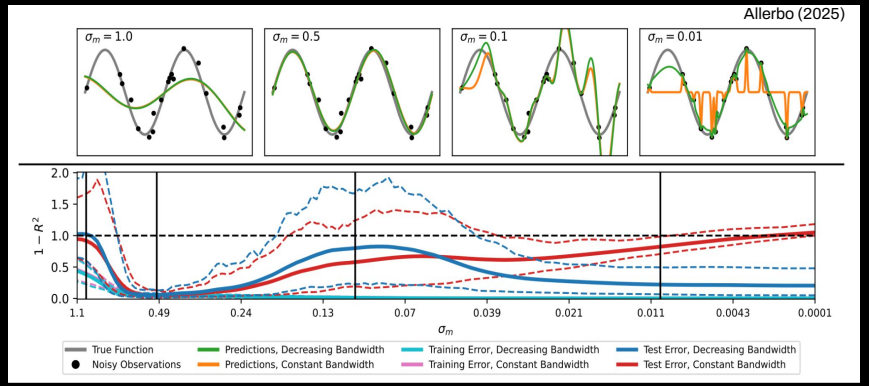
Pros: Computationally efficient and allows for high model complexity to observe double descent past interpolation point

Cons: RFF not learned from the data but instead intuited from random subsets, limiting the actual depth achievable in the second descent.

LITERATURE OVERVIEW

(CONT.)

Kernel Ridge Regression



Pro: Theoretically explains why constant-bandwidth KRR cannot double-descend — directly validates our experimental results.

Con: Proposed fix (KGD-D) requires iterative kernel recomputation, making it impractical for standard KRR pipelines.

Neural Network

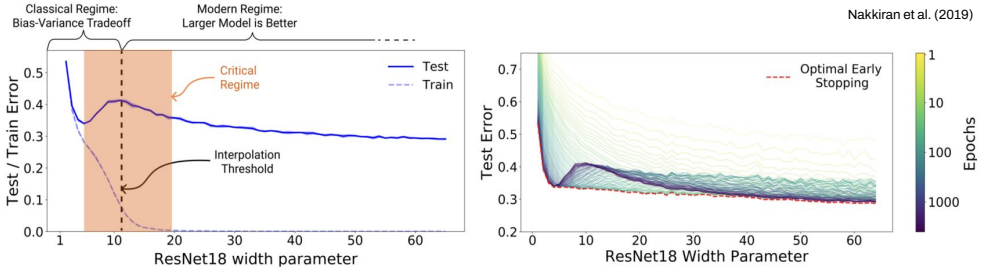


Figure 1: **Left:** Train and test error as a function of model size, for ResNet18s of varying width on CIFAR-10 with 15% label noise. **Right:** Test error, shown for varying train epochs. All models trained using Adam for 4K epochs. The largest model (width 64) corresponds to standard ResNet18.

Pros: Empirical validation cross multiple architectures. Shows double descent across model size, dataset size, and epochs.

Cons: Primarily classification tasks.

SUMMARY OF THE DATA

n_{train} = 500, n_{test} = 1000, NOISE = 1.0

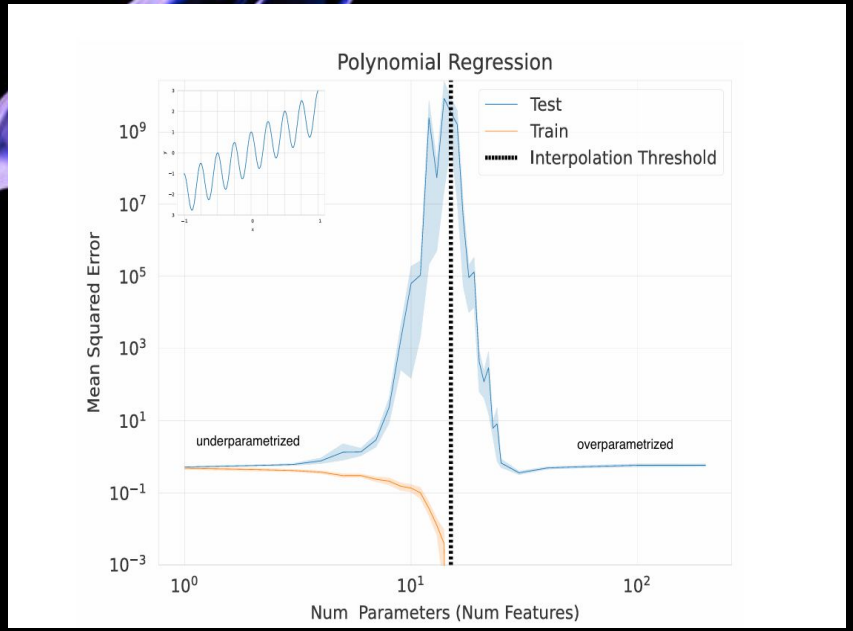
Synthetic Datasets: Friedman-1, 2, 3

Friedman-1	10+ Input Features (5 informative, 5+ noise) $$f(\mathbf{x}) = 10sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + N(0, \sigma)$$
Friedman-2	4+ Input Features (4 informative, 0+ noise) $$f(\mathbf{x}) = (x^2_0 + (x_1x_2 - x_1x_3)^{-1})^2)^{0.5} + N(0, \sigma)$$
Friedman-3	4+ Input Features (4 informative, 0+ noise) $$f(\mathbf{x}) = \arctan(x_1x_2 - (x_1x_3)^{-1}) / x_0) + N(0, \sigma)$$

METHODS

METHOD 1:

POLYNOMIAL REGRESSION



- Extension of linear regression where we first transform the input features into polynomial combinations and then fit a straight line through that transformed space.

- **Architecture**
  - Friedman1: Degree-3 polynomial expansion of 10 inputs
  - Friedman2/3: Degree-7 polynomial expansion of 4 inputs
  - Optional ridge regularization (L2)
  - No regularization (OLS)
  - Model complexity controlled by limiting features - Interpolation threshold at p = n_train = 100

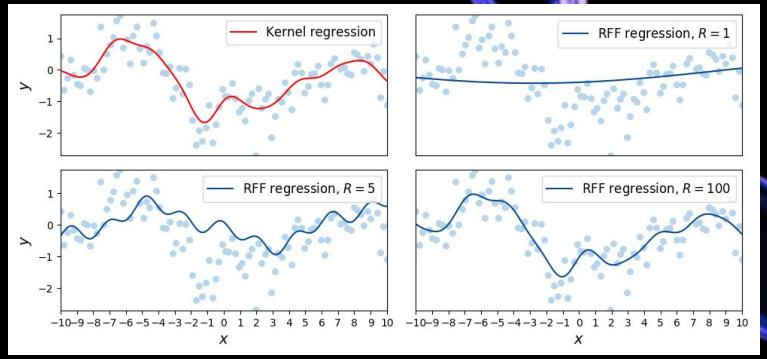
- Arxiv paper: Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle

METHOD 2:

RANDOM FEATURE REGRESSION

Random Fourier Feature Regression

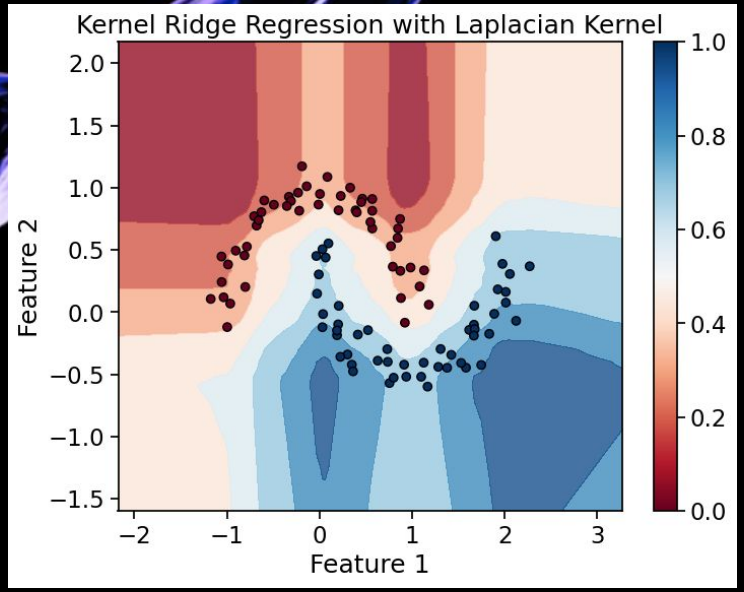
- Faster than kernel regression using a random finite set of features
- Approximates kernel method through the inner product of chosen features
- Random Fourier Features (RFF) specifically approximate the kernel through Fourier transforms and Monte-Carlo Averaging
- Friedman-1/2/3: Same architecture with parameters tied to kernel_dim = complexity; std_kernel = 0.5 (fixed bandwidth)
- ArXiv paper: *The generalization error of random features regression: Precise asymptotics and double descent curve*



The size of the random features set affect the trade-off between bias and variance.

METHOD 3:

KERNEL RIDGE REGRESSION



- KRR is a generalization of Linear Regression
  - Non-linear in data but linear in parameters
- Uses a kernel function (RBF) to measure similarity between data points — predictions are weighted combinations of these similarities
- Solves $(K + \alpha I)^{-1}y$, where α controls regularization
- Swept α from ~0.8 to ~10^{-8} (log-spaced) with 20 runs per α and 15% label corruption per run

Arxiv paper: *Changing the Kernel During Training Leads to Double Descent in Kernel Regression*

METHOD 4:

NEURAL NETWORK

Loss Function

$$L(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Forward Pass

$$\hat{y} = W_2 \cdot ReLU(W_1 x + b_1) + b_2$$

Optimization

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta)$$

where $\eta = 10^{-3}$ is the learning rate,
and $\theta = \{W_1, b_1, W_2, b_2\}$ are the trainable parameters

Architecture

- Single hidden layer MLP with ReLU activation
- Parameter count: p = h(d+2) + 1, where h = hidden units, d = input dimension
- Interpolation threshold at h* = ⌊n / (d + 2)⌋, where n = training samples

Training

- Full-batch gradient descent
- Adam optimizer, lr = 10^{-3}, epochs = 10,000
- He initialization for weights

Complexity Sweep

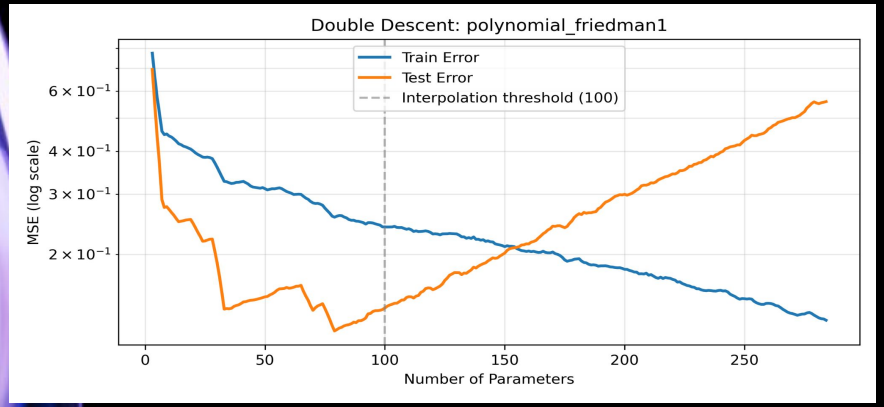
- Vary h from 1 to 400 hidden units
- Fine resolution around the interpolation threshold
- N_RUNS = 20 independent runs per complexity, averaged to smooth out stochastic variation

RESULTS

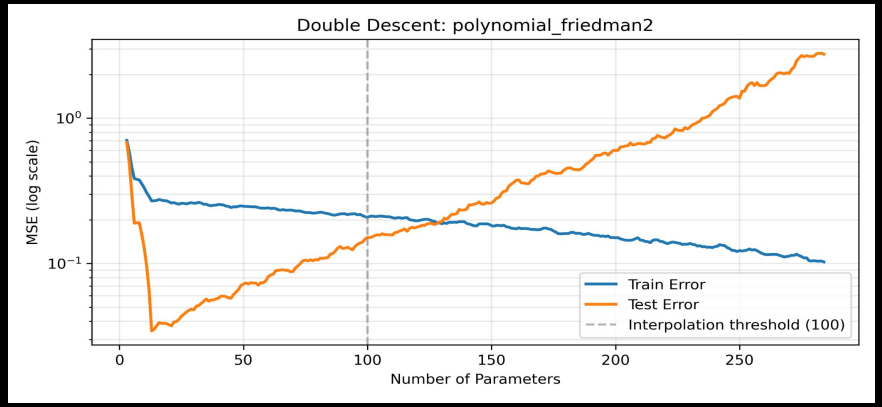
RESULTS

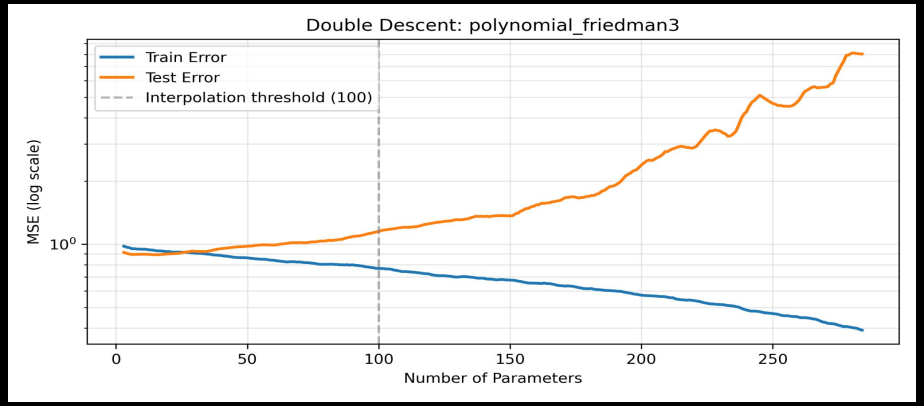
POLYNOMIAL REGRESSION

Min train error: 0.0691
MIn test error: 0.0162



Min train error: 0.2386
Min test error: 0.0285



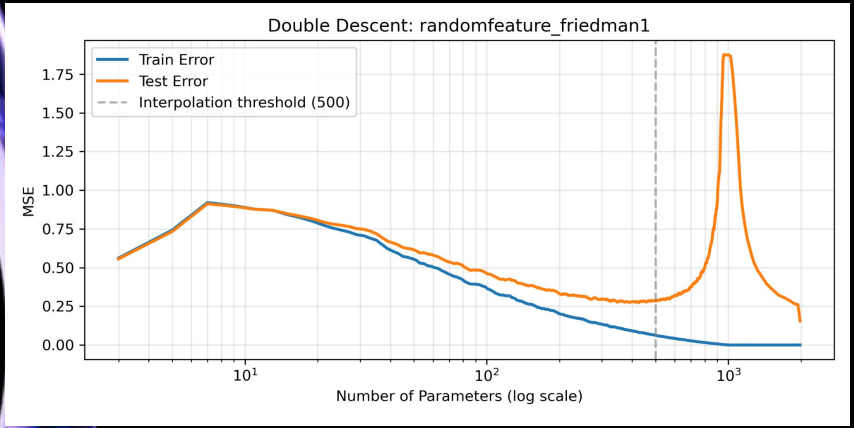


Min train error: 0.8833
Min test error: 0.8904

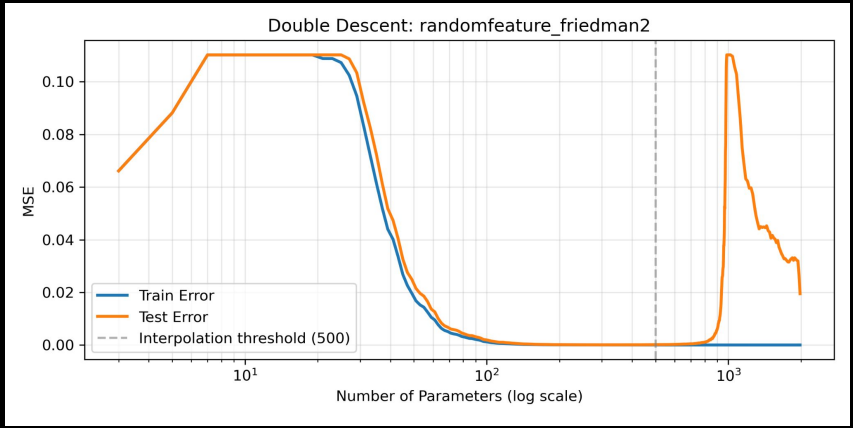
RESULTS

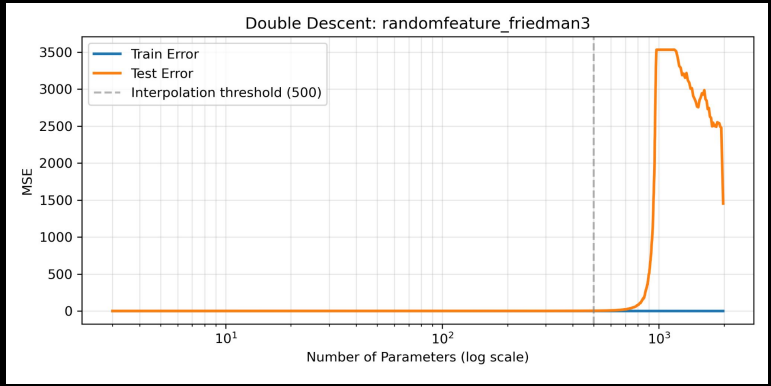
RANDOM FEATURE REGRESSION

Min train error: 0.0000
Min test error: 0.2559



Min train error: 0.0000
Min test error: 0.0001



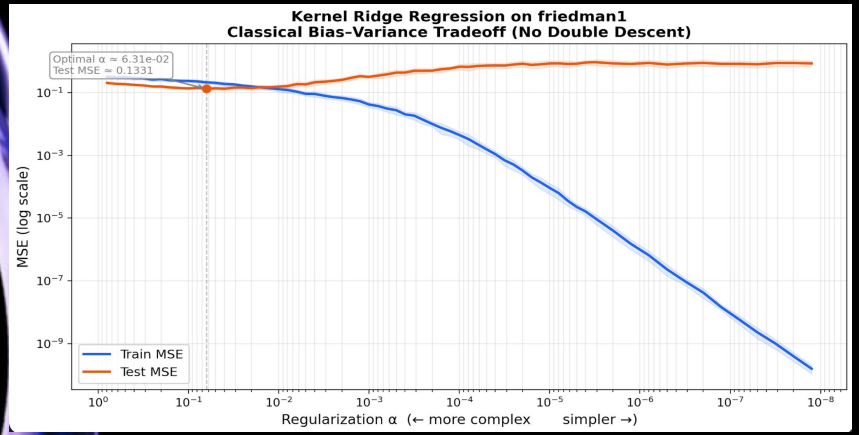


Min train error: 0.0000
Min test error: 0.8934

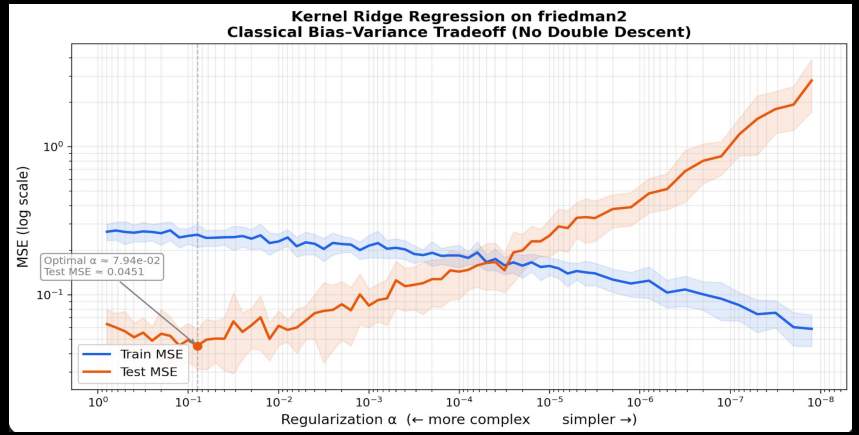
RESULTS

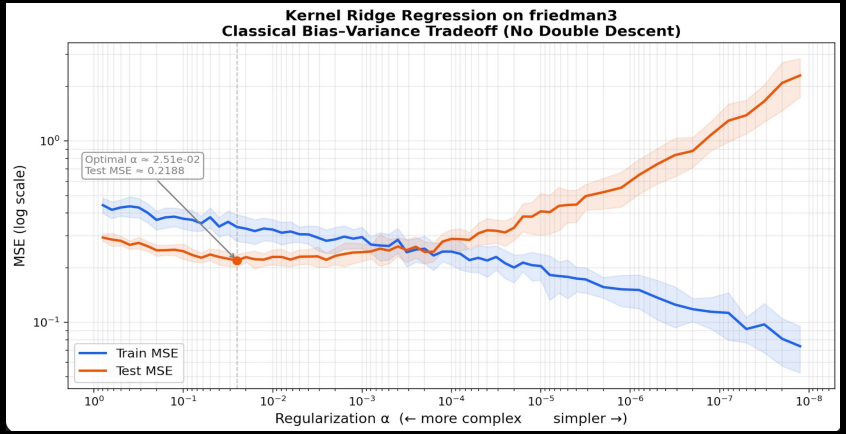
KERNEL RIDGE REGRESSION

Min train error: 0.0000
Min test error: 0.1331



Min train error: 0.0588
Min test error: 0.0451



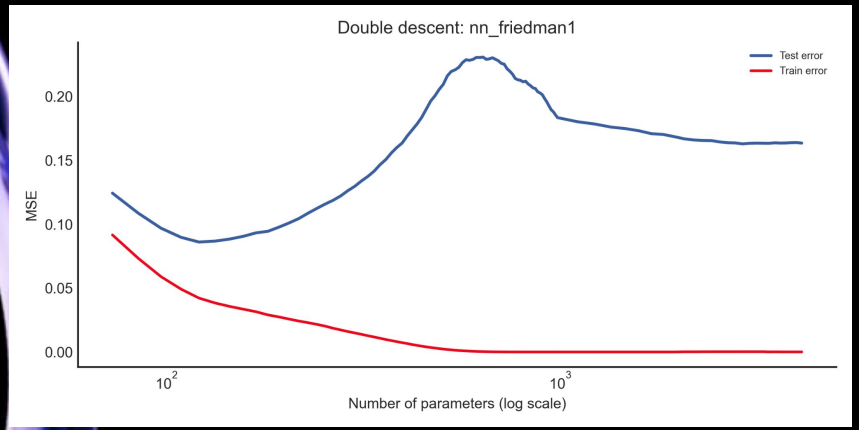


Min train error: 0.0738
Min test error: 0.2188

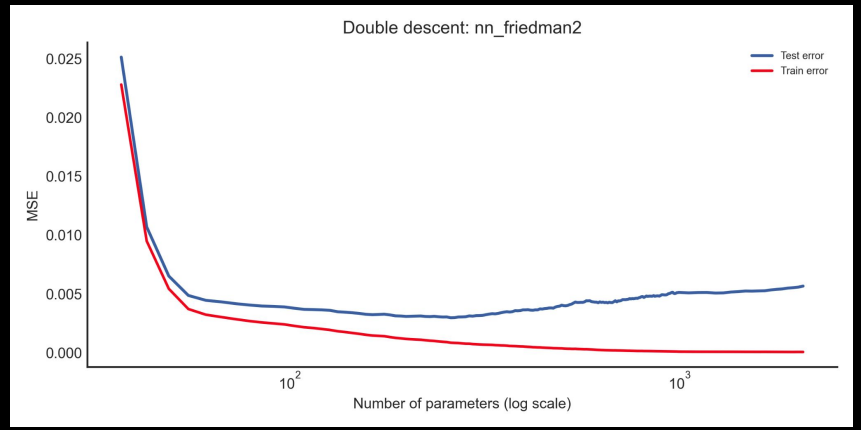
RESULTS

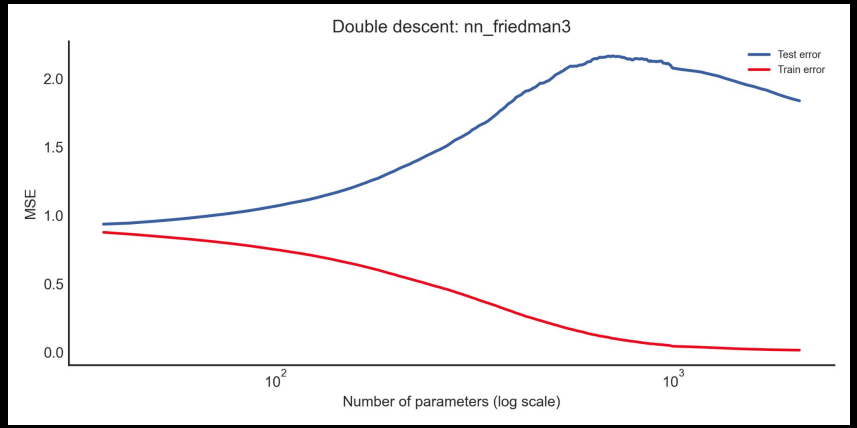
NEURAL NETWORK

Min train error: 0.0000
Min test error: 0.0773



Min train error: 0.0000
Min test error: 0.0025





Min train error: 0.0140
Min test error: 0.8987

DISCUSSION

What we learned

- Double descent is model and dataset-dependent.
  - Random Feature Regression and NN - Friedman 1
- Friedman 3 was the hardest dataset.
- Interpolation threshold aligns with theory.

Future Work

- Investigate Friedman 2 and Friedman 3 with Random Feature Regression and NN - more compute
- Implement decreasing kernel bandwidth like in Allerbo (2025)
- Try ensemble methods
- Extend NN Complexity and try different architectures.

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1. Allerbo, O. (2025). “Changing the kernel during training leads to double descent in kernel regression.” arXiv:2311.01762, 2025.
2. Friedman, J. H. (1991). Multivariate adaptive regression splines. The annals of statistics, 19(1), 1-67.
3. Mei, S., Andrea M. (2019). “The generalization error of random features regression: Precise asymptotics and double descent curve.” arXiv:1908.05355.
4. Nakkiran, P., Kaplun, G., Bansal, Y., Yang, T., Barak, B., & Sutskever, I. (2021). Deep double descent: Where bigger models and more data hurt. Journal of Statistical Mechanics: Theory and Experiment, 2021(12), 124003.
5. Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle, (March, 2023)